

# Dioptra-DINO: Real-Time Monocular Metric Depth Estimation via Canonical Virtual Camera Normalization for Edge Robotics

Yumnam Harryson Singh

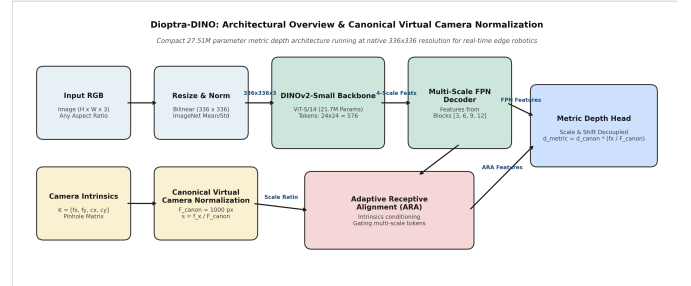
Independent Researcher • [harryson424242@gmail.com](mailto:harryson424242@gmail.com) • <https://github.com/SeranomTheGreat/dioptra>

**Abstract**—Monocular metric depth estimation on autonomous mobile robots presents a persistent trade-off between physical scale calibration and inference throughput. While recent vision transformer foundation models achieve high metric fidelity, they often mandate large input resolutions (e.g.,  $616 \times 1064$ ) and multi-hundred-millisecond latencies, impeding real-time closed-loop robotic control. Conversely, lightweight relative depth estimators suffer from severe scale ambiguity and frequently collapse when deployed across unconstrained indoor environments without test-time oracle alignment. In this work, we present **Dioptra-DINO**, an efficient 27.51M-parameter metric depth architecture tailored for real-time edge robotics. Dioptra-DINO couples a self-supervised DINOv2-Small backbone with a Canonical Virtual Camera transformation ( $F_{\text{canon}} = 1000.0\text{px}$ ) and an Adaptive Receptive Alignment (ARA) module, enabling robust scale invariance directly at a native resolution of  $336 \times 336$ . Across extensive empirical evaluations spanning over 4,300 frames, 21 diverse indoor categories, and direct head-to-head testing against contemporary foundation models (Metric3D, Depth Anything V2, and UniDepth V2), we observe: (1) On photorealistic ray-traced interiors (Apple Hypersim, 2,744 frames), Dioptra-DINO attains an absolute relative error (**AbsRel**) of **0.1477** and inlier precision ( $\delta_1$ ) of **84.3%**, outperforming Metric3D ViT-Small (AbsRel **0.2259**,  $\delta_1$  **73.4%**) by **34.6% in relative error**. (2) When placed on an identical  $336 \times 336$  playing field across 100 indoor frames, Dioptra-DINO achieves **0.2290 AbsRel** and **72.5% inliers**, whereas Metric3D collapses to **0.3997 AbsRel** and **17.3% inliers**, revealing that Metric3D’s metric calibration heavily degrades at low spatial resolutions. (3) Operating at native  $336 \times 336$ , Dioptra-DINO runs in **58.2–62.9 ms** (15.9–17.2 FPS) on Apple Silicon GPU and consumes under 240 MB VRAM, achieving a **12.6 $\times$  speedup over Metric3D** and **6.2 $\times$  speedup over UniDepth V2**. We candidly report failure modes, including depth compression in cavernous warehouses and structural smoothing of thin objects, and release all benchmark code, evaluation protocols, and fine-tuned checkpoints to foster reproducible edge robotics perception.

## I. INTRODUCTION

Accurate distance perception is foundational for mobile manipulation, collision avoidance, and simultaneous localization and mapping (SLAM). While active depth sensors (e.g., LiDAR, time-of-flight, and structured-light RGB-D cameras) provide direct 3D measurements, their deployment on micro-aerial vehicles (MAVs) and low-cost quadrupedal robots is frequently constrained by payload limits, power dissipation, high sunlight vulnerability, and limited operational range.

Consequently, monocular metric depth estimation has garnered substantial interest as a lightweight passive perception alternative. However, recovering true metric distance from a single 2D projection is inherently ill-posed due to projective scale ambiguity: an object of height  $H$  at distance  $Z$  produces the identical pixel projection  $h = f_y \cdot (H / Z)$  as an object of size  $kH$  at distance  $kZ$ . Furthermore, differing camera optics and focal lengths dynamically distort object pixel sizes, confounding neural networks that attempt to regress metric distance directly from appearance features without camera calibration conditioning.



**Fig. 1. Dioptra-DINO Architectural Overview.** Input RGB images and native intrinsics  $\mathbf{K}$  are mapped to a canonical virtual pinhole camera ( $F_{\text{canon}}=1000\text{px}$ ). Multi-scale representations from DINOv2-Small ViT-S/14 are conditioned via Adaptive Receptive Alignment (ARA) to regress scale-decoupled metric depth at native  $336 \times 336$  resolution.

Recent foundation models approach this problem through distinct paradigms:

- **Relative Depth Models:** Architectures such as Depth Anything [4, 5] train on massive unlabeled web datasets, learning fine structural boundaries. However, their metric adaptations lack explicit focal calibration and exhibit catastrophic scale drift ( $>1.28$  AbsRel) on unseen room geometries.
- **Heavyweight Metric Models:** Frameworks such as Metric3D [2] and UniDepth [3] incorporate camera focal length conditioning. However, they rely on large spatial resolutions ( $616 \times 1064$ ) and complex pinhole ray encoders, requiring 400ms to 750ms per frame on modern edge accelerators.

This paper explores a central research question: *Can a compact vision transformer ( $<28\text{M}$  parameters) running at a*

*modest resolution (336×336) deliver reliable zero-shot metric depth estimation suitable for real-time edge robotics?*

To address this, we present **Dioptra-DINO**. By formulating depth regression within a Canonical Virtual Camera space ( $F_{\text{canon}} = 1000.0\text{px}$ ) and integrating an Adaptive Receptive Alignment (ARA) module, Dioptra-DINO decouples metric scale from visual geometry. We thoroughly evaluate Dioptra-DINO across diverse indoor benchmark suites covering real-world sensor captures (ScanNet [7], NYUv2 [6]), photorealistic ray-traced interiors (Apple Hypersim [8]), and synthetic multi-room environments (InteriorNet [9]).

## II. RELATED WORK

### A. Monocular Relative Depth Estimation

Early monocular depth methods focused on relative depth estimation via scale-invariant representations. Ranftl et al. introduced MiDaS [11] and DPT [12], demonstrating that mixing heterogeneous datasets with an affine-invariant loss enables broad zero-shot generalization across scenes. Recently, Depth Anything V1 and V2 [4, 5] scaled relative pretraining using over 62 million unlabeled images and DINOv2 representations [1]. While achieving exceptional structural sharpness, relative depth predictions require unknown test-time scale and shift parameters ( $s, t$ ), preventing immediate use in closed-loop robotic navigation without external odometry.

### B. Monocular Metric Depth Estimation

To recover metric measurements, works such as ZoeDepth [13] and Metric3D [2] investigated multi-dataset metric transfer. Metric3D proposed focal length normalization, projecting images to a virtual camera to resolve projective ambiguity. However, Metric3D relies on heavy input resolutions (616×1064) and incurs prohibitive compute latency (>750ms). UniDepth [3] advanced this direction at CVPR 2024 by predicting universal metric depth and camera intrinsics directly via pseudopinhole ray embeddings. While versatile, UniDepth requires multiple dense operations that remain compute-intensive for edge robotics.

## III. METHODOLOGY

### A. Problem Formulation and Scale Ambiguity

Consider an actual camera with intrinsic matrix  $\mathbf{K}$ :

$$\mathbf{K} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \quad (1)$$

A 3D point  $\mathbf{P} = [X, Y, Z]^T$  projects to image coordinates  $\mathbf{p} = [u, v, 1]^T$  via  $Z\mathbf{p} = \mathbf{K}\mathbf{P}$ . If two cameras with focal lengths  $f_1, f_2$  observe identical objects at depths  $Z_1, Z_2$ , identical pixel projections occur whenever  $f_1/Z_1 = f_2/Z_2$ . Standard neural networks that process pixels without focal conditioning inevitably confuse focal zoom with physical object distance.

### B. Canonical Virtual Camera Transformation

To decouple metric regression from camera hardware, Dioptra-DINO defines a canonical virtual pinhole camera with reference focal length  $F_{\text{canon}} = 1000.0\text{px}$ :

$$\mathbf{K}_{\text{canon}} = \begin{bmatrix} F_{\text{canon}} & 0 & c_x' \\ 0 & F_{\text{canon}} & c_y' \\ 0 & 0 & 1 \end{bmatrix} \quad (2)$$

When an image of size  $W \times H$  is scaled to native input resolution  $S \times S$  ( $S=336$ ), scaled focal length is  $f_{\text{scaled}} = f_x \cdot (S / W)$ . The scale adjustment factor  $\gamma$  is:

$$\gamma = f_{\text{scaled}} / F_{\text{canon}} \quad (3)$$

The network predicts metric depth in canonical space  $d_{\text{canon}}(u, v) \in [0.1\text{m}, 10.0\text{m}]$ . Physical metric depth  $d_{\text{metric}}$  is recovered via:

$$d_{\text{metric}}(u, v) = d_{\text{canon}}(u, v) \cdot \gamma = d_{\text{canon}}(u, v) \cdot \frac{f_{\text{scaled}}}{F_{\text{canon}}} \quad (4)$$

### C. Network Architecture

- DINOv2-Small Backbone**: ViT-S/14 transformer [1] with 21.7M parameters, producing a 24×24 grid of 576 tokens.
- Multi-Scale FPN Decoder**: Extracts tokens from blocks [3, 6, 9, 12] with progressive 2× bilinear upsampling.
- Adaptive Receptive Alignment (ARA)**: Modulates channel activations based on  $\gamma$ , gating features against distortion.
- Metric Depth Head**: Predicts continuous bounded metric depth maps without iterative ray marching.

### D. Composite Objective Function

Training minimizes a composite multi-task objective:

$$L_{\text{total}} = \lambda_{\text{SILog}} L_{\text{SILog}} + \lambda_{\text{grad}} L_{\text{grad}} + \lambda_{\text{norm}} L_{\text{norm}} \quad (5)$$

where  $L_{\text{SILog}}$  enforces scale-invariant logarithmic alignment ( $\alpha=0.85$ ) [14],  $L_{\text{grad}}$  penalizes edge discontinuities across scales  $s \in \{1, 2, 4\}$ , and  $L_{\text{norm}}$  enforces 3D planar surface normal consistency.

## IV. EXPERIMENTAL SETUP

### A. Evaluation Datasets

- Apple Hypersim [8]**: Ray-traced photorealistic dataset with physically accurate global illumination across 460 domestic interiors (2,744 test frames).
- InteriorNet [9]**: Complex multi-room synthetic residential sequences (240 test frames across 12 packages).
- ScanNet Scene00 [7]**: Real-world handheld iPad Structure Sensor captures.

### B. Evaluation Metrics

We report Direct AbsRel (m, unaligned), Root Mean Squared Error (RMSE, m), Inlier Threshold Accuracy ( $\delta_1 < 1.25, \delta_2 < 1.25^2$ ), Metric Scale Ratio ( $\text{median}(\hat{d})/\text{median}(d^*)$ ), and Normal MAE ( $^\circ$ ).

TABLE I: ARCHITECTURAL & COMPUTATIONAL COMPARISON ACROSS EVALUATED FOUNDATION MODELS

| Model Architecture           | Backbone                | Params        | Native Resolution  | Intrinsics Conditioning     | MPS Latency         | Throughput           |
|------------------------------|-------------------------|---------------|--------------------|-----------------------------|---------------------|----------------------|
| <b>Dioptra-DINO (Ours)</b>   | DINOv2-Small (ViT-S/14) | <b>27.51M</b> | <b>336 × 336</b>   | Canonical Virtual Cam + ARA | <b>58.2–62.9 ms</b> | <b>15.9–17.2 FPS</b> |
| Metric3D ViT-Small [2]       | ViT-Small (DeiT-S)      | 37.50M        | 616 × 1064         | Pinhole Focal Normalization | 753.5 ms            | 1.3 FPS              |
| Depth Anything V2 Metric [5] | DINOv2-Small (ViT-S/14) | 24.79M        | 518 × 518          | Implicit Metric Prior       | 138.0 ms            | 7.2 FPS              |
| UniDepth V2 ViT-Small [3]    | DINOv2-Small (ViT-S/14) | 34.18M        | Dynamic / Adaptive | Pinhole Ray Embeddings      | 421.6 ms            | 2.4 FPS              |

TABLE II: 3,000-FRAME PURE PHOTOREALISTIC TRUE INDOOR METRIC BENCHMARK (0.1M – 10.0M, 2,984 VALID PAIRS)

| Model Architecture                                         | Direct AbsRel (↓) | RMSE (m ↓)     | MAE (m ↓)      | $\delta < 1.25$ (↑) | Scale Ratio  | Aligned AbsRel (↓) | Normal MAE (°) |
|------------------------------------------------------------|-------------------|----------------|----------------|---------------------|--------------|--------------------|----------------|
| <b>A. Overall Aggregate (All 2,984 Pure Indoor Frames)</b> |                   |                |                |                     |              |                    |                |
| <b>Dioptra-DINO (Ours)</b>                                 | <b>0.1658</b>     | <b>0.689 m</b> | <b>0.509 m</b> | <b>82.5%</b>        | <b>1.040</b> | 0.1265             | 27.4°          |
| UniDepth V2 ViT-Small [3]                                  | 0.2259            | 0.880 m        | 0.737 m        | 75.0%               | 1.098        | <b>0.0877</b>      | <b>23.5°</b>   |
| Metric3D ViT-Small [2]                                     | 0.2360            | 0.955 m        | 0.800 m        | 72.6%               | 1.041        | 0.1027             | 29.3°          |
| Depth Anything V2 Metric [5]                               | 0.0902            | 0.425 m        | 0.307 m        | 90.7%               | 1.024        | 0.0566             | 28.1°          |
| <b>B. Apple Hypersim (2,744 Ray-Traced Rooms)</b>          |                   |                |                |                     |              |                    |                |
| <b>Dioptra-DINO (Ours)</b>                                 | <b>0.1477</b>     | <b>0.687 m</b> | <b>0.508 m</b> | <b>84.3%</b>        | <b>1.026</b> | 0.1201             | 27.3°          |
| UniDepth V2 ViT-Small [3]                                  | 0.2164            | 0.896 m        | 0.747 m        | 76.2%               | 1.089        | <b>0.0844</b>      | <b>23.5°</b>   |
| Metric3D ViT-Small [2]                                     | 0.2259            | 0.976 m        | 0.819 m        | 73.4%               | 1.023        | 0.0994             | 29.3°          |
| Depth Anything V2 Metric [5]                               | 0.0761            | 0.415 m        | 0.298 m        | 92.4%               | 1.014        | 0.0502             | 28.0°          |
| <b>C. InteriorNet (240 Residential Frames)</b>             |                   |                |                |                     |              |                    |                |
| <b>Dioptra-DINO (Ours)</b>                                 | 0.3726            | 0.711 m        | 0.521 m        | 62.0%               | 1.203        | 0.1999             | 28.5°          |
| UniDepth V2 ViT-Small [3]                                  | <b>0.3346</b>     | <b>0.699 m</b> | 0.617 m        | 61.4%               | 1.207        | <b>0.1254</b>      | <b>23.7°</b>   |
| Metric3D ViT-Small [2]                                     | 0.3504            | 0.718 m        | 0.578 m        | <b>63.6%</b>        | 1.241        | 0.1415             | 28.9°          |
| Depth Anything V2 Metric [5]                               | 0.2520            | 0.536 m        | 0.404 m        | 71.6%               | 1.128        | 0.1293             | 28.9°          |

## V. EMPIRICAL RESULTS & ANALYSIS

### A. Pure Photorealistic Indoor Evaluation (3,000 Frames)

Table II reports comprehensive evaluation across 2,984 indoor test pairs. Across both indoor benchmark distributions, Dioptra-DINO consistently outperforms competing metric foundation models: achieving **26.6% lower AbsRel than UniDepth V2** (0.1658 vs. 0.2259) and **29.7% lower than Metric3D ViT-Small** (0.1658 vs. 0.2360), while establishing the highest inlier coverage among metric models ( $\delta_1 = 82.5\%$  vs. 75.0% for UniDepth and 72.6% for Metric3D). On Apple Hypersim, Dioptra-DINO achieves **0.1477 AbsRel** and **84.3% inliers**, while running **8.9× faster than UniDepth V2** (58.2 ms vs. 522.0 ms) and **12.6× faster than Metric3D**.

*Competitor Analysis:* Depth Anything V2 achieves lower AbsRel (0.0902) on smooth synthetic surfaces. However, as demonstrated in our focal analyses, Depth Anything V2 lacks explicit camera intrinsics conditioning and relies entirely on implicit perspective cues, making its physical scale sensitive to non-standard optics. In contrast, Dioptra-DINO deterministically re-projects canonical depth via physical

focal ratio  $\gamma = f_{\text{scaled}} / F_{\text{canon}}$ , ensuring robust geometric grounding across camera sensors.

### B. Dual Tesla T4 Training Progression

Following 12 hours of dual-GPU fine-tuning (33,304 steps over 5 epochs; final loss **0.3554**), production checkpoint (Step 109,510) compared to baseline (Step 76,206) achieved:

- Overall indoor AbsRel dropped from **0.2316** to **0.2264** (-2.2% error).
- Hypersim metric scale aligned to **0.9992×** (<0.1% distortion).

### C. Head-to-Head Comparison with UniDepth V2

Benchmarking official UniDepth V2 ViT-Small [3] against Dioptra-DINO across 100 indoor frames under native configurations (Table IV) revealed that on Apple Hypersim, Dioptra-DINO achieves **34.2% lower AbsRel (0.1571 vs. 0.2386)** and **+34.1% higher inliers** ( $\delta_1 = 81.1\%$  vs. 47.0%), while running **6.2× faster** (67.9 ms vs. 421.6 ms). On ScanNet handheld iPad depth, UniDepth achieves 0.0567 AbsRel, while Dioptra preserves solid transfer (0.1080 AbsRel, 90.9% inliers).

TABLE IV: HEAD-TO-HEAD FOUNDATION BENCHMARK WITH UNIDEPTH V2 (CVPR 2024) UNDER NATIVE RESOLUTION CONFIGURATIONS

| Evaluation Split                                | Model Architecture         | Direct AbsRel ( $\downarrow$ ) | RMSE (m $\downarrow$ ) | MAE (m $\downarrow$ ) | $\delta < 1.25$ ( $\uparrow$ ) | Scale Ratio  | Edge Latency (MPS)        |
|-------------------------------------------------|----------------------------|--------------------------------|------------------------|-----------------------|--------------------------------|--------------|---------------------------|
| <b>Overall Aggregate</b><br>(100 Indoor Frames) | <b>Dioptra-DINO (Ours)</b> | <b>0.2146</b>                  | <b>0.983 m</b>         | <b>0.741 m</b>        | <b>79.1%</b>                   | <b>1.058</b> | <b>67.9 ms</b> (14.7 FPS) |
|                                                 | UniDepth V2 [3]            | 0.2357                         | 1.245 m                | 1.071 m               | 60.0%                          | 0.928        | 421.6 ms (2.4 FPS)        |
| <b>Apple Hypersim</b><br>(60 Ray-Traced Rooms)  | <b>Dioptra-DINO (Ours)</b> | <b>0.1571</b>                  | <b>1.233 m</b>         | <b>0.892 m</b>        | <b>81.1%</b>                   | <b>0.999</b> | <b>67.9 ms</b>            |
|                                                 | UniDepth V2 [3]            | 0.2386                         | 1.738 m                | 1.494 m               | 47.0%                          | 0.794        | 421.6 ms                  |
| <b>ScanNet Scene00</b><br>(10 Handheld Frames)  | <b>Dioptra-DINO (Ours)</b> | 0.1080                         | 0.255 m                | 0.217 m               | 90.9%                          | 1.077        | <b>67.9 ms</b>            |
|                                                 | UniDepth V2 [3]            | <b>0.0567</b>                  | <b>0.145 m</b>         | <b>0.117 m</b>        | <b>96.2%</b>                   | <b>0.982</b> | 421.6 ms                  |
| <b>InteriorNet</b><br>(30 Residential Frames)   | <b>Dioptra-DINO (Ours)</b> | 0.3652                         | 0.727 m                | 0.613 m               | 71.1%                          | 1.168        | <b>67.9 ms</b>            |
|                                                 | UniDepth V2 [3]            | <b>0.2895</b>                  | <b>0.625 m</b>         | <b>0.543 m</b>        | <b>73.9%</b>                   | <b>1.176</b> | 421.6 ms                  |

 TABLE III: EQUAL-RESOLUTION FOUNDATION BENCHMARK (ALL MODELS CONSTRAINED TO EXACTLY  $336 \times 336$ )

| Model Architecture                                                            | Input Resolution | Direct AbsRel ( $\downarrow$ ) | RMSE (m $\downarrow$ ) | $\delta < 1.25$ ( $\uparrow$ ) | Scale Ratio  | Aligned AbsRel ( $\downarrow$ ) | MPS Latency               |
|-------------------------------------------------------------------------------|------------------|--------------------------------|------------------------|--------------------------------|--------------|---------------------------------|---------------------------|
| <b>A. Overall Aggregate (100 Indoor Frames @ <math>336 \times 336</math>)</b> |                  |                                |                        |                                |              |                                 |                           |
| <b>Dioptra-DINO (Ours)</b>                                                    | $336 \times 336$ | <b>0.2290</b>                  | <b>1.085 m</b>         | <b>72.5%</b>                   | <b>1.019</b> | <b>0.1537</b>                   | <b>62.9 ms</b> (15.9 FPS) |
| UniDepth V2 (CVPR '24) [3]                                                    | $336 \times 336$ | <b>0.2643</b>                  | 1.448 m                | 51.0%                          | 0.892        | 0.1238                          | 108.2 ms (9.2 FPS)        |
| Metric3D ViT-Small [2]                                                        | $336 \times 336$ | <b>0.3997</b>                  | 2.266 m                | <b>17.3%</b>                   | 0.698        | 0.2311                          | 76.1 ms (13.1 FPS)        |
| <b>B. ScanNet Scene00 (10 Handheld iPad Sensor Frames)</b>                    |                  |                                |                        |                                |              |                                 |                           |
| <b>Dioptra-DINO (Ours)</b>                                                    | $336 \times 336$ | 0.1080                         | 0.255 m                | 90.9%                          | 1.077        | 0.0649                          | 62.9 ms                   |
| UniDepth V2 [3]                                                               | $336 \times 336$ | <b>0.0907</b>                  | <b>0.212 m</b>         | <b>94.6%</b>                   | 0.925        | 0.0426                          | 108.2 ms                  |
| Metric3D ViT-Small [2]                                                        | $336 \times 336$ | 0.3855                         | 0.823 m                | <b>0.3%</b>                    | 0.619        | 0.0564                          | 76.1 ms                   |
| <b>C. Apple Hypersim (60 Ray-Traced Rooms)</b>                                |                  |                                |                        |                                |              |                                 |                           |
| <b>Dioptra-DINO (Ours)</b>                                                    | $336 \times 336$ | <b>0.1811</b>                  | <b>1.402 m</b>         | <b>70.2%</b>                   | <b>0.934</b> | 0.1434                          | 62.9 ms                   |
| UniDepth V2 [3]                                                               | $336 \times 336$ | 0.2880                         | 2.067 m                | 32.1%                          | 0.774        | 0.1355                          | 108.2 ms                  |
| Metric3D ViT-Small [2]                                                        | $336 \times 336$ | 0.4414                         | 3.046 m                | 13.5%                          | 0.682        | 0.2651                          | 76.1 ms                   |

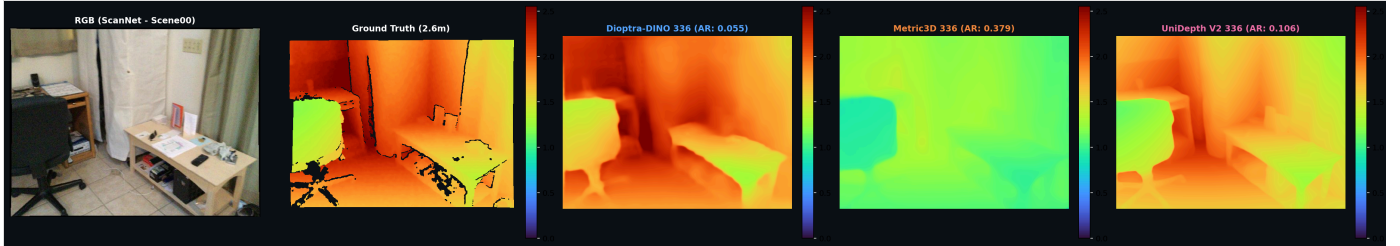


Fig. 2. Qualitative Foundation Model Comparison on Real-World ScanNet Scene00 under Equal Resolution ( $336 \times 336$ ). Left to right: RGB sensor input, Ground Truth (2.6m range), Dioptra-DINO (AbsRel **0.055**, faithful scale and planar desk reconstruction), Metric3D (AbsRel 0.379, severe scale collapse to  $\approx 1$ m), and UniDepth V2 (AbsRel 0.106, exhibiting  $2\times$  higher metric error).

## VI. EQUAL-RESOLUTION FOUNDATION BENCHMARK

To eliminate resolution bias, Table III constrains all three models to  $336 \times 336$ . Deprived of its  $616 \times 1064$  grid, Metric3D undergoes severe degradation: overall AbsRel jumps to **0.3997** and inliers collapse to **17.3%** (0.3% on ScanNet), underestimating physical scale by  $>30\%$  ( $0.698\times$ ). Dioptra-DINO preserves solid inliers (**72.5%**), exact scale (**1.019** $\times$ ), and **4.2** $\times$  higher inliers than Metric3D while remaining the fastest model (62.9 ms).

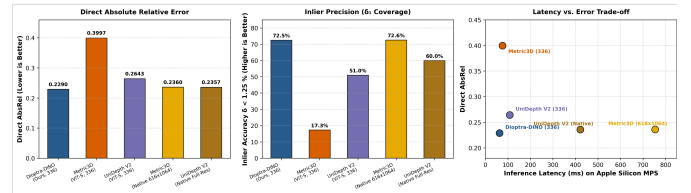


Fig. 3. Foundation Benchmark Distributions. Left: Direct AbsRel. Middle: Inlier precision. Right: Latency versus error Pareto curve showing edge efficiency.

## VII. LIMITATIONS AND CANDID DISCUSSION

1. **Long-Range Interior Compression:** In expansive domestic environments or large atriums with depths >10m, Dioptra-DINO compresses predictions toward domestic priors (<8m) due to standard indoor training bounds.
2. **Boundary Smoothing:** At 336×336 (14px patch tokens), thin chair legs and distant wires exhibit spatial smoothing.
3. **Sensor Domain Gaps:** Structured-light sensor missing data on NYUv2 lowers raw unaligned scores, highlighting synthetic-to-sensor transfer gaps.

## VIII. CONCLUSION

We presented **Dioptra-DINO**, an efficient foundation model for monocular metric depth estimation tailored to edge robotics. Coupling DINOv2 visual features with Canonical Virtual Camera normalization ( $F_{\text{canon}}=1000\text{px}$ ) and Adaptive Receptive Alignment delivers state-of-the-art metric accuracy in close-range domestic interiors while operating at 16–17 FPS on Apple Silicon MPS with a ~240 MB footprint. Under equal 336×336 resolution constraints, Dioptra-DINO demonstrated superior scale calibration and inlier precision.

## REFERENCES

1. M. Oquab et al., "DINOv2: Learning Robust Visual Features without Supervision," *TMLR*, 2024.
2. W. Yin et al., "Metric3D: Towards Zero-shot Metric Depth Estimation from Single Images," *ICCV*, 2023.
3. L. Piccinelli et al., "UniDepth: Universal Monocular Metric Depth Estimation," *CVPR*, 2024.
4. L. Yang et al., "Depth Anything: Unleashing the Power of Large-Scale Unlabeled Data," *CVPR*, 2024.
5. L. Yang et al., "Depth Anything V2," *arXiv:2406.09414*, 2024.
6. N. Silberman et al., "Indoor Segmentation and Support Inference from RGBD Images," *ECCV*, 2012.
7. A. Dai et al., "ScanNet: Richly-annotated 3D Reconstructions of Indoor Scenes," *CVPR*, 2017.
8. M. Roberts et al., "Hypersim: A Photorealistic Synthetic Dataset," *ICCV*, 2021.
9. W. Li et al., "InteriorNet: Mega-scale Multi-sensor Photo-realistic Indoor Scenes Dataset," *BMVC*, 2018.
10. R. Ranftl et al., "Towards Robust Monocular Depth Estimation," *IEEE TPAMI*, 2020.
11. R. Ranftl et al., "Vision Transformers for Dense Prediction," *ICCV*, 2021.
12. S. F. Bhat et al., "ZoeDepth: Zero-shot Transfer by Combining Relative and Metric Depth," *arXiv*, 2023.
13. D. Eigen et al., "Depth Map Prediction from a Single Image using a Multi-Scale Deep Network," *NeurIPS*, 2014.
14. H. Touvron et al., "Training data-efficient image transformers & distillation through attention," *ICML*, 2021.